

# Deep Learning for teens

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# Embeddings

- ▶ Supervised learning
- ▶ Selfsupervised learning
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- ▶ Embedding capture similarities between words

# Embeddings

- ▶ Supervised learning:
  - ▶ (no major embedding algorithm)
- ▶ Self-supervised learning:
  - ▶ Word2Vec
  - ▶ FastText
- ▶ Unsupervised learning:
  - ▶ GloVe

# GloVe

- ▶ GloVe builds a co-occurrence matrix: how often words appear together across a large corpus.
- ▶ It then factorizes this matrix to produce dense vector representations.
- ▶ The key insight: ratios of co-occurrence probabilities encode semantic meaning.
- ▶ Example: ice is more likely to co-occur with solid than with gas, while steam is more likely to co-occur with gas than with solid.
- ▶ These relationships get captured in the vector space.

# GloVe

- ▶ Global + Local context: It combines the statistical power of matrix factorization (global corpus information) with the efficiency of prediction models (local context).
- ▶ Pretrained embeddings: GloVe provides large pretrained vectors (trained on Wikipedia, Common Crawl, etc.), which are widely used in NLP tasks.
- ▶ Semantic relationships: Captures analogies like king - man + woman  $\approx$  queen, similar to Word2Vec.

# GloVe vs Word2Vec

- ▶ **\*\*In short:\*\***
  - ▶ - **\*\*Word2Vec\*\*** learns embeddings by **\*\*predicting context words\*\*** using a neural network (local context).
  - ▶ - **\*\*GloVe\*\*** learns embeddings by **\*\*factorizing a co-occurrence matrix\*\*** of the whole corpus (global statistics).
- ▶ ☞ Word2Vec = prediction-based, local context.
- ▶ ☞ GloVe = count-based, global co-occurrence.
- ▶ ✓ **Key difference in practice:**
  - ▶ With **Word2Vec**, you train embeddings directly on your corpus.
  - ▶ With **GloVe**, you usually load **pretrained embeddings** (trained on massive corpora like Wikipedia or Common Crawl) and use them in your models.

# FastText

- ▶ **Subword information:** Instead of treating each word as a single token, FastText breaks words into **character n-grams**.
  - ▶ An n-gram is a substring of length n taken from a word. FastText breaks each word into many such character n-grams so it can learn subword patterns.
- ▶ Word2Vec: learns whole-word embeddings.
- ▶ GloVe: learns whole-word embeddings from global co-occurrence.
- ▶ FastText: learns embeddings from subword pieces, making it robust to rare and unseen words.



# supervised learning- supervised learning and self-supervised learning

- ▶ So, **supervised learning**—we don't really have a direct example of that for embedding algorithms. They're not learned that way.
- ▶ **Unsupervised learning**, perfect example: that's **GloVe**. It learns from co-occurrence statistics, completely unsupervised, no labels at all.
- ▶ And for **self-supervised learning**, that's like **Word2Vec** and of course **FastText**—those models create their own labels from the data by predicting context words. So there you have it! Let me know if that fits what you're looking for.

# Self-supervised learning

- ▶ means the model creates its own labels from the data.
- ▶ It uses part of the input to predict another part of the input.

# Unsupervised vs Self-supervised

- ▶ In unsupervised learning, we're just grouping or categorizing data into clusters—no labels at all, just patterns.
- ▶ unsupervised is just clustering, and self-supervised is making its own labels from the data.

# Homework

- Use the Sentiment140 dataset (first 10,000 tweets), apply Word2Vec, GloVe, and FastText on the cleaned text, and then compare which model shows the best similar words.